ENERGY SECTOR — SENIOR QUANT RESEARCH

CEG · VST · CCJ — Benchmark, Factor, Optimization & Monte-Carlo Framework

Independent Power Producers & Nuclear Fuel — AI/Data-Center Power Demand Thematic

Report date:	2026-10-01
Coverage window:	Jan-2025 – Sep-2026 (trailing), fwd view to FY2027
Prepared for:	xristos.katsambas@gmail.com
Classification:	Internal research workbook — not a solicitation

WHAT'S NEW IN THIS VERSION (vs. the original workbook)
✓ XLE sector benchmark added — alpha, tracking error, information ratio, up/down capture
✓ Macro factor sensitivities — single-factor betas to WTI crude, Henry Hub gas, uranium spot, 10Y yield
✓ Real running-peak max drawdown (replaces the MIN/MAX proxy) + historical VaR/CVaR (95%/99%) + skew/kurtosis
✓ Sortino ratio (downside deviation) alongside Sharpe
✓ Mean-variance portfolio optimization — 66-point efficient frontier, max-Sharpe & min-vol portfolios identified
✓ Monte Carlo simulation (10,000 paths, 12-month horizon) — percentile fan chart, terminal-value histogram, simulation-based VaR/CVaR
✓ Companion Python repo (src/) for full reproducibility: fetch_data.py, monte_carlo.py, build_workbook.py

WORKBOOK MAP	
1. Cover	This page — scope, what's new, disclaimers

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2. Market Data	Monthly levels: CEG/VST/CCJ/XLE + WTI, Henry Hub, 10Y yield, uranium spot — sourced
3. Returns & Risk	Returns, real drawdown, Sharpe/Sortino, VaR/CVaR, skew/kurtosis — live formulas
4. Benchmark & Factors	Alpha/beta/TE/IR/capture vs. XLE; univariate betas to oil/gas/uranium/rates
5. Portfolio Optimization	66-point efficient frontier, max-Sharpe & min-variance portfolios, CML
6. Monte Carlo	10,000-path simulation: percentile fan, terminal histogram, sim VaR/CVaR
7-9. Deep Dives	CEG / VST / CCJ — fundamentals, technicals, catalysts, earnings history
10. Catalyst Calendar	Consolidated dated events, Oct-2026–Q1-2027
11. Recommendation	Optimizer + simulation-informed sizing, scenario framing, disclaimer

METHODOLOGY & LIMITATIONS (read before using any number in this workbook): Monthly observation frequency (21 months) is a THIN sample for statistical inference — covariance, VaR, skew/kurtosis and regression betas all carry wide estimation error at n=20. Max drawdown, VaR/CVaR and correlation use month-end data only and will UNDERSTATE true intra-month peak-to-trough risk. Macro factor betas are univariate (one factor at a time) — true multi-factor attribution requires controlling for collinearity among oil/gas/uranium/rates, which is a job for proper regression software, not single-variable Excel formulas; treat these as directional, not precise. The Monte Carlo simulation assumes i.i.d. multivariate-normal monthly returns — it does not capture fat tails, regime shifts, or autocorrelation, and is a shape-of-uncertainty illustration, not a forecast. WTI and uranium monthly series are analyst-reconstructed approximations (see Market Data sourcing note); XLE, Henry Hub, and 10Y yield are sourced from public data providers. This is NOT investment advice or a broker-dealer research report. Verify every figure against a live terminal before acting.

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MONTHLY MARKET DATA — Prices, Benchmark & Macro Factors (Jan-2025 – Sep-2026)								
Month	CEG	VST	CCJ	XLE	WTI	Henry Hub	UST 10Y	Uranium
Jan-25	$296.99	$166.66	$49.34	$41.83	73.00	4.13	4.63	73.00
Feb-25	$248.05	$132.57	$43.95	$43.43	71.00	4.19	4.45	69.50
Mar-25	$199.99	$116.68	$41.08	$44.95	68.50	4.12	4.28	66.00
Apr-25	$221.63	$128.79	$45.06	$38.72	63.00	3.42	4.28	64.50
May-25	$304.08	$159.54	$58.42	$39.22	61.50	3.12	4.42	68.00
Jun-25	$320.58	$192.80	$74.09	$41.12	65.00	3.02	4.38	71.50
Jul-25	$345.49	$207.46	$74.78	$42.29	67.00	3.20	4.39	73.00
Aug-25	$306.27	$188.13	$77.24	$43.83	64.50	2.91	4.26	75.50
Sep-25	$327.24	$195.11	$83.70	$43.69	63.50	2.97	4.12	78.00
Oct-25	$374.90	$187.52	$102.01	$43.10	61.00	3.19	4.06	80.50
Nov-25	$362.75	$178.12	$88.34	$44.23	59.50	3.79	4.09	79.00
Dec-25	$351.71	$160.89	$91.49	$44.10	58.00	4.26	4.14	81.00
Jan-26	$279.44	$157.92	$123.39	$50.36	57.00	7.72	4.21	83.50
Feb-26	$328.42	$173.42	$118.40	$55.16	58.50	3.62	4.13	82.00
Mar-26	$278.39	$150.12	$108.61	$60.82	60.00	3.04	4.25	80.50
Apr-26	$312.04	$157.62	$123.04	$59.22	59.00	2.77	4.32	84.00
May-26	$287.31	$160.01	$112.70	$55.89	57.50	2.94	4.48	86.50
Jun-26	$247.99	$158.63	$101.86	$53.11	56.00	3.15	4.47	85.00
Jul-26	$262.35	$148.19	$86.38	$59.55	58.00	2.89	4.60	87.00
Aug-26	$274.77	$137.37	$98.76	$63.96	60.50	2.78	4.68	88.50
Sep-26	$254.02	$138.35	$86.67	$61.54	62.00	2.85	4.98	89.45
Spot (2026-10	**264.58**	**139.05**	**87.39**	**61.80**	**62.00**	**2.85**	**4.98**	**89.45**

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Sources: CEG/VST/CCJ — digrin.com, StockAnalysis.com, Yahoo Finance, Robinhood (retrieved 2026-10-01). XLE — digrin.com monthly closes. Henry Hub natural gas spot ($/MMBtu) — U.S. EIA (eia.gov/dnav/ng), Sep-2026 value estimated (not yet published at report date). UST 10Y — multpl.com monthly averages, cross-checked against market commentary for Sep-2026. Uranium spot (U3O8, $/lb) — Sep-2026 value sourced from Trading Economics; Jan-2025–Aug-2026 are ANALYST-RECONSTRUCTED APPROXIMATIONS consistent with the publicly reported direction of the uranium market (IMF/NUEXCO series not independently reproducible here) — replace with a UxC/TradeTech subscription feed before relying on these for the Uranium factor column. WTI crude ($/bbl) is similarly an ANALYST-RECONSTRUCTED APPROXIMATION based on EIA STEO commentary (direct EIA/FRED time-series endpoints were not reachable from this environment) — replace with a live feed (EIA, Bloomberg, FactSet) before using the Oil factor column for anything beyond directional illustration.

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RETURNS & RISK METRICS — live formulas, linked to Market Data

Month	CEG	VST	CCJ	XLE		WTI %chg	HH %chg	UST10Y Δ	Uranium %chg	CEG dwn²	VST dwn²	CCJ dwn²
Feb-25	-16.48%	-20.45%	-10.92%	3.83%		-2.74%	1.45%	-0.18	-4.79%	2.715%	4.184%	1.193%
Mar-25	-19.38%	-11.99%	-6.53%	3.50%		-3.52%	-1.67%	-0.17	-5.04%	3.754%	1.437%	0.426%
Apr-25	10.82%	10.38%	9.69%	-13.86%		-8.03%	-16.99%	0.00	-2.27%	0.000%	0.000%	0.000%
May-25	37.20%	23.88%	29.65%	1.29%		-2.38%	-8.77%	0.14	5.43%	0.000%	0.000%	0.000%
Jun-25	5.43%	20.85%	26.82%	4.84%		5.69%	-3.21%	-0.04	5.15%	0.000%	0.000%	0.000%
Jul-25	7.77%	7.60%	0.93%	2.85%		3.08%	5.96%	0.01	2.10%	0.000%	0.000%	0.000%
Aug-25	-11.35%	-9.32%	3.29%	3.64%		-3.73%	-9.06%	-0.13	3.42%	1.289%	0.868%	0.000%
Sep-25	6.85%	3.71%	8.36%	-0.32%		-1.55%	2.06%	-0.14	3.31%	0.000%	0.000%	0.000%
Oct-25	14.56%	-3.89%	21.88%	-1.35%		-3.94%	7.41%	-0.06	3.21%	0.000%	0.151%	0.000%
Nov-25	-3.24%	-5.01%	-13.40%	2.62%		-2.46%	18.81%	0.03	-1.86%	0.105%	0.251%	1.796%
Dec-25	-3.04%	-9.67%	3.57%	-0.29%		-2.52%	12.40%	0.05	2.53%	0.093%	0.936%	0.000%
Jan-26	-20.55%	-1.85%	34.87%	14.20%		-1.72%	81.22%	0.07	3.09%	4.222%	0.034%	0.000%
Feb-26	17.53%	9.82%	-4.04%	9.53%		2.63%	-53.11%	-0.08	-1.80%	0.000%	0.000%	0.164%
Mar-26	-15.23%	-13.44%	-8.27%	10.26%		2.56%	-16.02%	0.12	-1.83%	2.321%	1.805%	0.684%
Apr-26	12.09%	5.00%	13.29%	-2.63%		-1.67%	-8.88%	0.07	4.35%	0.000%	0.000%	0.000%
May-26	-7.93%	1.52%	-8.40%	-5.62%		-2.54%	6.14%	0.16	2.98%	0.628%	0.000%	0.706%
Jun-26	-13.69%	-0.86%	-9.62%	-4.97%		-2.61%	7.14%	-0.01	-1.73%	1.873%	0.007%	0.925%
Jul-26	5.79%	-6.58%	-15.20%	12.13%		3.57%	-8.25%	0.13	2.35%	0.000%	0.433%	2.310%
Aug-26	4.73%	-7.30%	14.33%	7.41%		4.31%	-3.81%	0.08	1.72%	0.000%	0.533%	0.000%
Sep-26	-7.55%	0.71%	-12.24%	-3.78%		2.48%	2.52%	0.30	1.07%	0.570%	0.000%	1.499%

DRAWDOWN DETAIL — running-peak method (real max drawdown, monthly granularity)

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Month	CEG Px	CEG RunMax	CEG DD	VST Px	VST RunMax	VST DD	CCJ Px	CCJ RunMax	CCJ DD
Jan-25	$296.99	$296.99	0.0%	$166.66	###	0.0%	$49.34	$49.34	0.0%
Feb-25	$248.05	$296.99	-16.5%	$132.57	###	-20.5%	$43.95	$49.34	-10.9%
Mar-25	$199.99	$296.99	-32.7%	$116.68	###	-30.0%	$41.08	$49.34	-16.7%
Apr-25	$221.63	$296.99	-25.4%	$128.79	###	-22.7%	$45.06	$49.34	-8.7%
May-25	$304.08	$304.08	0.0%	$159.54	###	-4.3%	$58.42	$58.42	0.0%
Jun-25	$320.58	$320.58	0.0%	$192.80	###	0.0%	$74.09	$74.09	0.0%
Jul-25	$345.49	$345.49	0.0%	$207.46	###	0.0%	$74.78	$74.78	0.0%
Aug-25	$306.27	$345.49	-11.4%	$188.13	###	-9.3%	$77.24	$77.24	0.0%
Sep-25	$327.24	$345.49	-5.3%	$195.11	###	-6.0%	$83.70	$83.70	0.0%
Oct-25	$374.90	$374.90	0.0%	$187.52	###	-9.6%	$102.01	$102.01	0.0%
Nov-25	$362.75	$374.90	-3.2%	$178.12	###	-14.1%	$88.34	$102.01	-13.4%
Dec-25	$351.71	$374.90	-6.2%	$160.89	###	-22.4%	$91.49	$102.01	-10.3%
Jan-26	$279.44	$374.90	-25.5%	$157.92	###	-23.9%	$123.39	$123.39	0.0%
Feb-26	$328.42	$374.90	-12.4%	$173.42	###	-16.4%	$118.40	$123.39	-4.0%
Mar-26	$278.39	$374.90	-25.7%	$150.12	###	-27.6%	$108.61	$123.39	-12.0%
Apr-26	$312.04	$374.90	-16.8%	$157.62	###	-24.0%	$123.04	$123.39	-0.3%
May-26	$287.31	$374.90	-23.4%	$160.01	###	-22.9%	$112.70	$123.39	-8.7%
Jun-26	$247.99	$374.90	-33.9%	$158.63	###	-23.5%	$101.86	$123.39	-17.4%
Jul-26	$262.35	$374.90	-30.0%	$148.19	###	-28.6%	$86.38	$123.39	-30.0%
Aug-26	$274.77	$374.90	-26.7%	$137.37	###	-33.8%	$98.76	$123.39	-20.0%
Sep-26	$254.02	$374.90	-32.2%	$138.35	###	-33.3%	$86.67	$123.39	-29.8%

SUMMARY RISK & RETURN STATISTICS				
Metric	CEG	VST	CCJ	XLE
Risk-free rate	4.3%			
Mean monthly	0.22%	-0.35%	3.90%	2.16%

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Std dev, montl	14.72%	11.17%	15.44%	6.72%
Annualized re	-9.0%	-10.6%	40.2%	26.1%
Annualized vo	51.0%	38.7%	53.5%	23.3%
Max drawdow	-33.9%	-33.8%	-30.0%	n/a
Skewness (mc	0.60	0.53	0.63	-0.29
Kurtosis (exce	0.43	0.20	-0.75	0.47
Historical VaR	-19.43%	-13.79%	-13.49%	-6.03%
Historical VaR	-20.33%	-19.12%	-14.86%	-12.29%
Historical CVa	-20.55%	-20.45%	-15.20%	-13.86%
Sortino ratio	-0.41	-0.59	1.49	n/a
Sharpe ratio (	-0.26	-0.38	0.67	0.94
Beta (vs. S&P	1.12	1.41	1.01	1.00

CORRELATION MATRIX (monthly returns, incl. benchmark)				
	CEG	**VST**	**CCJ**	**XLE**
CEG	1.00	0.72	0.41	-0.18
VST	0.72	1.00	0.52	-0.25
CCJ	0.41	0.52	1.00	0.07
XLE	-0.18	-0.25	0.07	1.00

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BENCHMARK & MACRO FACTOR ANALYSIS — single-factor regressions, linked to Returns & Risk

TABLE A — RELATIVE TO XLE (SECTOR BENCHMARK)

Ticker	Beta (XLE)	R²	Alpha (ann., Jensen)	Tracking Error (ann.)	Info Ratio	Up Capture	Down Capture
CEG	-0.39	0.03	-4.9%	##	-0.55	-0.10	-0.37
VST	-0.42	0.06	-5.7%	##	-0.69	-0.18	-0.21
CCJ	0.17	0.01	32.2%	##	0.24	0.68	-0.81

TABLE B — MACRO FACTOR SENSITIVITIES (univariate SLOPE/R², one factor at a time)

Ticker	β Oil (WTI)	R²	β Nat Gas (HH)	R²	β Uranium	R²	β Rates (10Y Δ)	R²
CEG	0.37	0.01	-0.28	##	2.21	0.23	0.16	0.02
VST	0.46	0.02	-0.09	##	1.84	0.27	0.19	0.05
CCJ	-0.25	0.00	0.21	##	3.04	0.39	-0.05	0.00

Reading this tab: Beta(XLE) > 1 means the stock amplifies sector moves; Alpha is the annualized excess return left over after stripping out that market-beta exposure (CAPM/Jensen's alpha) — a positive alpha means the stock did better than its sector-beta alone would predict. Information Ratio is alpha per unit of tracking error — the standard buy-side measure of whether active risk is being paid for. Up/Down Capture above 1.0 in Up and below 1.0 in Down is the textbook 'asymmetric' profile a long-only holder wants. Table B betas are univariate (each factor regressed alone) because oil, gas, uranium and rates are themselves correlated, so a true multi-factor model would need to control for that collinearity — treat these as directional signals of sensitivity, not precise, independent exposures.

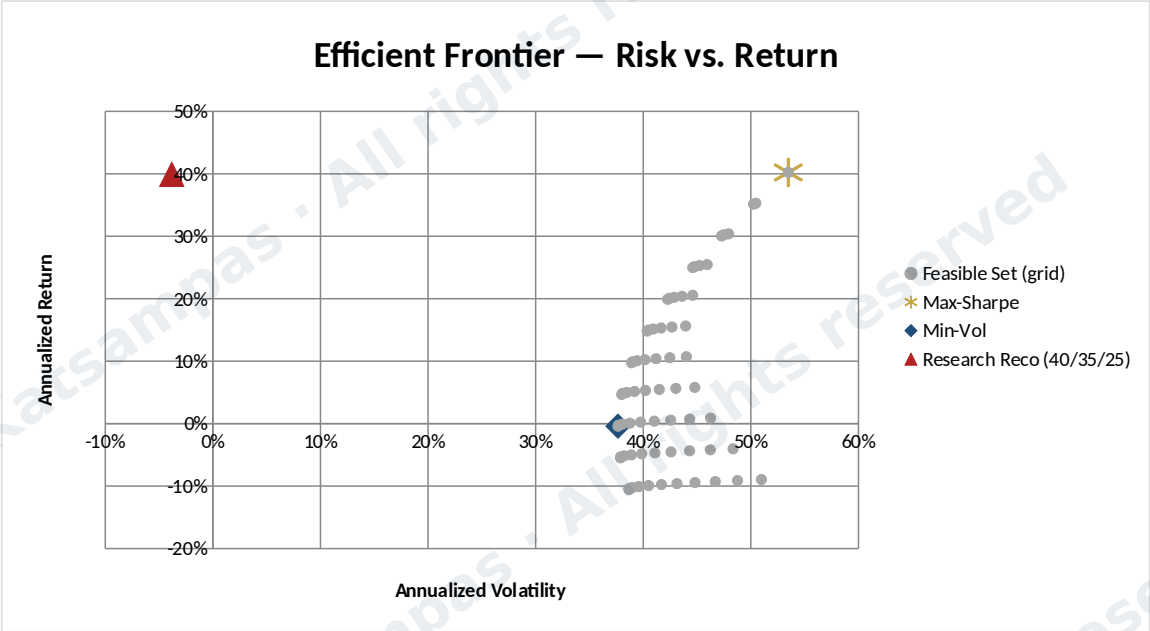
CEG-XLE	VST-XLE	CCJ-XLE
-20.30%	-24.28%	-14.75%
-22.88%	-15.49%	-10.03%
24.68%	24.24%	23.55%
35.91%	22.58%	28.36%
0.58%	16.00%	21.98%
4.92%	4.76%	-1.91%
-14.99%	-12.96%	-0.35%
7.17%	4.03%	8.68%
15.91%	-2.54%	23.23%
-5.86%	-7.63%	-16.02%
-2.75%	-9.38%	3.86%
-34.74%	-16.04%	20.67%
-6.34%	-18.71%	-27.32%
-2.67%	-14.71%	6.93%
-3.77%	4.50%	-8.46%

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PORTFOLIO OPTIMIZATION — Mean-Variance Efficient Frontier (CEG / VST / CCJ)

MONTHLY COVARIANCE MATRIX (sample, n-1)			
	CEG	VST	CCJ
CEG	0.02166	0.01190	0.00943
VST	0.01190	0.01247	0.00898
CCJ	0.00943	0.00898	0.02384

EFFICIENT-FRONTIER GRID (10% weight steps) + NAMED PORTFOLIOS					
w(CEG)	w(VST)	w(CCJ)	Port. Return	Port. Vol	Sharpe
0%	0%	100%	40.2%	53.5%	0.67
0%	10%	90%	35.1%	50.3%	0.61
0%	20%	80%	30.1%	47.3%	0.54
0%	30%	70%	25.0%	44.6%	0.46
0%	40%	60%	19.9%	42.3%	0.37
0%	50%	50%	14.8%	40.4%	0.26
0%	60%	40%	9.7%	38.9%	0.14
0%	70%	30%	4.7%	38.0%	0.01
0%	80%	20%	-0.4%	37.6%	-0.13
0%	90%	10%	-5.5%	37.9%	-0.26
0%	100%	0%	-10.6%	38.7%	-0.38
10%	0%	90%	35.3%	50.5%	0.61
10%	10%	80%	30.2%	47.5%	0.55
10%	20%	70%	25.1%	44.8%	0.47
10%	30%	60%	20.1%	42.4%	0.37
10%	40%	50%	15.0%	40.5%	0.26
10%	50%	40%	9.9%	39.0%	0.14
10%	60%	30%	4.8%	38.1%	0.01
10%	70%	20%	-0.2%	37.7%	-0.12
10%	80%	10%	-5.3%	37.9%	-0.25
10%	90%	0%	-10.4%	38.7%	-0.38
20%	0%	80%	30.4%	47.9%	0.54
20%	10%	70%	25.3%	45.2%	0.46
20%	20%	60%	20.2%	42.9%	0.37
20%	30%	50%	15.1%	40.9%	0.27
20%	40%	40%	10.1%	39.4%	0.15
20%	50%	30%	5.0%	38.5%	0.02
20%	60%	20%	-0.1%	38.1%	-0.12



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20%	70%	10%	-5.2%	38.2%	-0.25
20%	80%	0%	-10.2%	39.0%	-0.37
30%	0%	70%	25.5%	45.9%	0.46
30%	10%	60%	20.4%	43.6%	0.37
30%	20%	50%	15.3%	41.7%	0.26
30%	30%	40%	10.2%	40.2%	0.15
30%	40%	30%	5.2%	39.2%	0.02
30%	50%	20%	0.1%	38.8%	-0.11
30%	60%	10%	-5.0%	38.9%	-0.24
30%	70%	0%	-10.1%	39.6%	-0.36
40%	0%	60%	20.6%	44.6%	0.36
40%	10%	50%	15.5%	42.7%	0.26
40%	20%	40%	10.4%	41.2%	0.15
40%	30%	30%	5.3%	40.2%	0.03
40%	40%	20%	0.2%	39.8%	-0.10
40%	50%	10%	-4.8%	39.8%	-0.23
40%	60%	0%	-9.9%	40.5%	-0.35
50%	0%	50%	15.6%	43.9%	0.26
50%	10%	40%	10.6%	42.5%	0.15
50%	20%	30%	5.5%	41.5%	0.03
50%	30%	20%	0.4%	41.0%	-0.10
50%	40%	10%	-4.7%	41.1%	-0.22
50%	50%	0%	-9.8%	41.7%	-0.34
60%	0%	40%	10.7%	44.0%	0.15
60%	10%	30%	5.6%	43.0%	0.03
60%	20%	20%	0.6%	42.5%	-0.09
60%	30%	10%	-4.5%	42.6%	-0.21
60%	40%	0%	-9.6%	43.1%	-0.32
70%	0%	30%	5.8%	44.8%	0.03
70%	10%	20%	0.7%	44.3%	-0.08
70%	20%	10%	-4.4%	44.3%	-0.20
70%	30%	0%	-9.4%	44.8%	-0.31
80%	0%	20%	0.9%	46.3%	-0.07
80%	10%	10%	-4.2%	46.2%	-0.18
80%	20%	0%	-9.3%	46.7%	-0.29
90%	0%	10%	-4.0%	48.3%	-0.17
90%	10%	0%	-9.1%	48.8%	-0.28
100%	0%	0%	-9.0%	51.0%	-0.26

NAMED REFERENCE PORTFOLIOS

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w(CEG)	w(VST)	w(CCJ)	Port. Return	Port. Vol	Sharpe	
33%	33%	33%	6.9%	39.8%	0.07	**Equal-Weight (33/33/33)**
40%	35%	25%	2.8%	39.9%	-0.04	**Research Reco (40/35/25)**

IDENTIFIED OPTIMAL PORTFOLIOS (from the grid above)						
Portfolio	**w(CEG)**	**w(VST)**	**w(CCJ)**	**Port. Return**	**Port. Vol**	**Sharpe**
Max-Sharpe	**0%**	**0%**	**100%**	**40.2%**	**53.5%**	**###**
Min-Volatility	**0%**	**80%**	**20%**	**-0.4%**	**37.6%**	**###**

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MONTE CARLO SIMULATION — 12-Month Portfolio Outcomes (10,000 paths)

Pre-computed by src/monte_carlo.py (seed=42, multivariate-normal monthly returns fit to the 20-month sample, 40/35/25 CEG/VST/CCJ weights). Re-run that script to refresh these values — they are NOT live Excel formulas. See Cover tab for method and limitations.

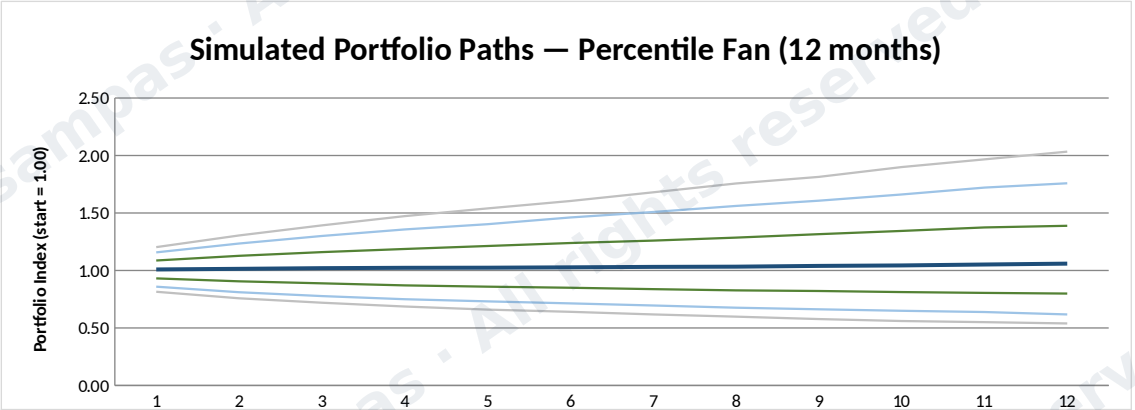
SIMULATION SUMMARY (terminal, 12-month)	
Simulations run	10000
Mean terminal retu	14.1%
Median terminal re	5.9%
Std dev, terminal r	47.5%
Simulated VaR 95	-46.1%
Simulated VaR 99	-59.5%
Simulated CVaR 9	-54.3%
Simulated CVaR 9	-64.1%
P(loss) over 12 m	44.3%
P(loss > 10%) ove	34.8%
P(gain > 25%) ove	34.2%

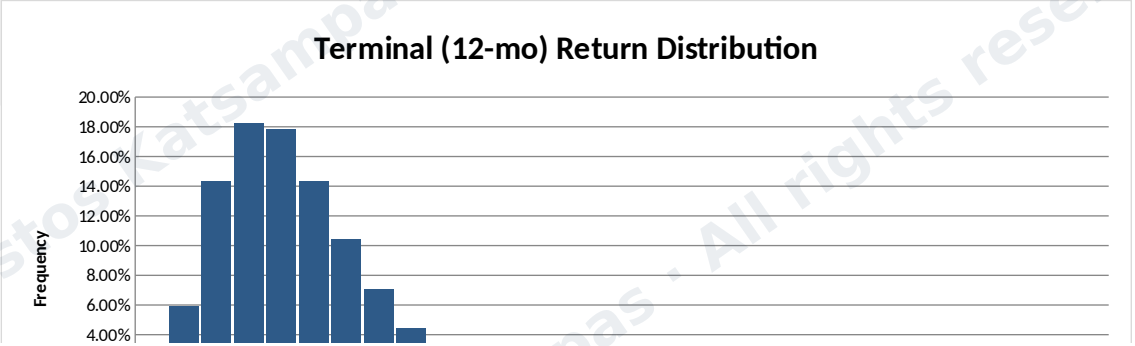
PERCENTILE FAN — PORTFOLIO INDEX PATH (start = 1.00)

MONTH	P5	P10	P25	P50	P75	P90	P95
1	0.81	0.86	0.93	1.01	1.09	1.16	1.20
2	0.76	0.81	0.91	1.01	1.13	1.23	1.30
3	0.72	0.78	0.89	1.02	1.16	1.30	1.39
4	0.69	0.75	0.87	1.02	1.19	1.36	1.47
5	0.66	0.73	0.86	1.02	1.21	1.40	1.54
6	0.64	0.71	0.85	1.03	1.24	1.46	1.60
7	0.62	0.70	0.84	1.03	1.26	1.51	1.68
8	0.60	0.68	0.83	1.03	1.29	1.56	1.76
9	0.58	0.66	0.82	1.04	1.32	1.61	1.81
10	0.56	0.65	0.81	1.04	1.34	1.66	1.90
11	0.55	0.64	0.80	1.05	1.37	1.72	1.97
12	0.54	0.62	0.80	1.06	1.39	1.76	2.03

TERMINAL RETURN DISTRIBUTION (histogram, 30 bins)

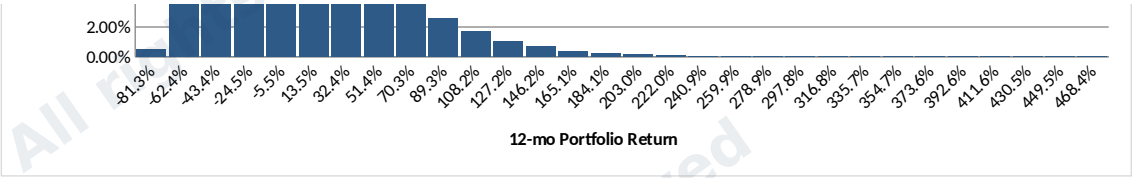
Bin (left edge)	Frequency
-81.3%	0.53%
-62.4%	5.94%
-43.4%	14.33%
-24.5%	18.21%
-5.5%	17.80%
13.5%	14.31%
32.4%	10.39%
51.4%	7.04%
70.3%	4.45%
89.3%	2.61%





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108.2%	1.69%
127.2%	1.02%
146.2%	0.70%
165.1%	0.37%
184.1%	0.26%
203.0%	0.17%
222.0%	0.09%
240.9%	0.04%
259.9%	0.02%
278.9%	0.00%
297.8%	0.00%
316.8%	0.01%
335.7%	0.01%
354.7%	0.00%
373.6%	0.00%
392.6%	0.00%
411.6%	0.00%
430.5%	0.00%
449.5%	0.00%
468.4%	0.01%



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CEG — Constellation Energy Corporation

Sector: Utilities — Independent Power Producers | Largest U.S. nuclear fleet operator (~31,676 MW) | Baltimore, MD

KEY FUNDAMENTALS		
Price (2026-10-01)	**$264.58**	
Market capitalization	**~$92.3B**	
52-week range	**$228.63 – $412.70**	
P/E ratio (TTM)	**25.5x**	
Forward P/E	**21.4x**	
EPS (TTM)	**$10.24 – $10.33**	
Revenue (TTM)	**$31.27B (+26.0% YoY)**	
Dividend yield	**0.64%**	
Beta (5Y monthly)	**1.12**	
FY2026 EPS guidance (raised)	**$11.50 – $12.50**	
Analyst consensus	**Buy / Strong Buy**	
Avg. 12-mo price target	**~$379 (BMO, Aug-2026)**	

TECHNICAL PICTURE		
Trend status	**Below 200-day SMA; lower half of 52-wk range**	
Price vs. 52-wk high/low	**-35.9% / +15.7%**	
1-year price return	**-20.3%**	
Key support / resistance	**~$250 support / ~$310–330 resistance**	

RECENT QUARTERLY EARNINGS HISTORY

Quarter	Report Date	EPS Est.	EPS Actual	Surprise	Price Reaction
Q2 2026	2026-08-06	$2.28	$2.55	+11.7%	Guidance raised to $11.50-12.50

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Q1 2026	2026-05-11	$2.60	$2.74	+5.3%	-1.3% (1-day)
Q4 2025	2026-02-24	N/A	$2.30	+2.4%	+6.4% (1-day)
Q3 2025	2025-11-07	$3.13	$3.04	-2.9%	+2.0% (1-day)

CATALYST CALENDAR & RECENT EVENTS (chronological)	
2026-08-06	Q2 2026 results; FY26 guidance raised to $11.50-12.50; 920 MW of new long-term PPAs signed.
2026 (ongoing)	FERC/NRC approvals advance Crane Clean Energy Center (former TMI Unit 1) restart for Microsoft.
2026 (ongoing)	Calpine acquisition ($16.4B) integration; divesting Brazos Valley + 4.4GW PJM assets to LS Power for DOJ clearance.
2026-11-09	Q3 2026 earnings (confirmed). Consensus EPS $3.65–$3.67.

BULL CASE

✓ Largest U.S. nuclear fleet = irreplaceable carbon-free baseload exactly as hyperscalers sign 20-yr PPAs for that profile.

✓ Guidance raised twice in 2026 even as the stock sits ~36% below its Oct-2025 high — a valuation reset, not a broken thesis.

BEAR CASE / RISKS

⚠️ Premium multiple (25x+ TTM) for a utility, even after the pullback.

⚠️ Large, complex Calpine integration with forced divestitures still in progress; PJM market-design intervention risk sector-wide.

Sources: SEC EDGAR 8-K, TipRanks, StockAnalysis.com, Yahoo Finance — retrieved 2026-10-01.

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VST — Vistra Corp.

Sector: Utilities — Independent Power Producers | Integrated retail + generation (~39GW) | Irving, TX

KEY FUNDAMENTALS		
Price (2026-10-01)	**$139.05**	
Market capitalization	**~$46.4B**	
52-week range	**$132.66 – $217.10**	
P/E ratio (TTM)	**21.5x – 23.8x**	
EPS (TTM)	**$6.00**	
Revenue (FY2025)	**$17.74B (+3.0% YoY)**	
Dividend yield	**0.65%**	
Beta (5Y monthly, reported)	**1.41 (cross-provider range 0.4–1.9)**	
FY2026 adj. EBITDA guidance	**$6.8B – $7.6B (reaffirmed)**	
Credit rating	**BBB- (Fitch upgrade, Mar-2026)**	
Analyst consensus	**Strong Buy**	
Avg. 12-mo price target	**~$217.58 (+54.5% implied)**	

TECHNICAL PICTURE		
Trend status	**Downtrend since Aug-2025 ($217) to ~$139**	
Price vs. 52-wk high/low	**-36.0% / +4.8%**	
1-year price return	**-32.4%**	
Key support / resistance	**~$132–138 support / ~$160–175 resistance**	

RECENT QUARTERLY EARNINGS HISTORY

Quarter	Report Date	EPS Est.	EPS Actual	Surprise	Price Reaction
Q2 2026	2026-08-07	N/A	N/A	N/A	TD adj. EBITDA $3.261B vs $2.589B PY

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Q1 2026	2026-05-11	N/A	N/A	Record Q1 EBITDA	+5% premarket
FY2025	2026-02 (approx.)	N/A	net i	-69.5% YoY	Weather/deal-related noise
Q3 2025	2025-11-06	N/A	N/A	N/A	-6.1% (1-day)

CATALYST CALENDAR & RECENT EVENTS (chronological)	
2026-01-12	20-year, 2.6 GW nuclear PPA with Meta — flagship zero-carbon long-duration contract.
2026-01-15	PJM market-intervention headline risk — shares fell ~9% on affordability-regulation reports.
2026-03	Fitch upgraded Vistra to investment grade (BBB-).
2026 H2 (pending)	Cogentrix gas portfolio acquisition (5,500 MW) targeted to close; not yet in guidance.
2026-11-06	Q3 2026 earnings (confirmed) — first quarter with potential Cogentrix commentary.

BULL CASE

✓ Purest merchant-power / direct-hyperscaler-PPA exposure of the three (Meta 2.6GW/20-yr deal).

✓ Investment-grade upgrade lowers financing cost during an active M&A phase; highest analyst conviction of the group (~55% implied upside).

BEAR CASE / RISKS

⚠️ Most exposed of the three to PJM/ERCOT regulatory intervention — already proven with a 9% single-day drop.

⚠️ Merchant power economics are inherently more volatile than regulated/contracted revenue; Cogentrix deal unclosed.

Sources: StockAnalysis.com, TradingView, Robinhood, Morningstar, PR Newswire — retrieved 2026-10-01.

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CCJ — Cameco Corporation

Sector: Energy — Uranium Mining & Nuclear Fuel Cycle | Uranium, Fuel Services, Westinghouse (49%) | Saskatoon, Canada

KEY FUNDAMENTALS	
Price (2026-10-01)	**$87.39**
Market capitalization	**~$37.8B**
52-week range	**$77.70 – $135.24**
P/E ratio (TTM)	**~147x–152x (thin TTM earnings)**
EPS (TTM)	**$0.57**
Revenue (FY2025)	**$3.48B (+11.0% YoY)**
Net income (FY2025)	**$589.6M (+243.1% YoY)**
Dividend yield	**0.20%**
Beta (5Y monthly)	**1.01**
Analyst consensus	**Buy / Moderate Buy**
Avg. 12-mo price target	**$127.77 – $148.68 (wide dispersion)**

TECHNICAL PICTURE	
Trend status	**Round-trip: $78 (Apr-25) → $135 (Oct-25/Jan-26) → ~$87 now**
Price vs. 52-wk high/low	**-35.4% / +12.5%**
1-year price return	**-4.2% (highly path-dependent)**
Key support / resistance	**~$86–90 support / ~$110–123 resistance**

RECENT QUARTERLY EARNINGS HISTORY

Quarter	Report Date	EPS Est.	EPS Actual	Surprise	Price Reaction
Q2 2026	2026-07-31	$0.26	$0.13	-50% to -64%	-2.1% to -3.2% (1-day)
Q1 2026	2026-05-05	$0.25–$0.36	22–$0	mixed	-3.2% on initial miss

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Q4 2025	2026-02-13	$0.32	33–$0	~0–+28.6%	N/A
Q3 2025	2025-11-05	$0.23	$0.05	-78.3%	N/A

CATALYST CALENDAR & RECENT EVENTS (chronological)	
2026-07-31	Q2 2026 EPS $0.13 vs $0.26 est. (miss); shares drifted +10.8% higher over the following 33 days.
2026 (ongoing)	Westinghouse (49%-owned) IPO speculation — potential embedded-value unlock.
2026-10-30 (est.)	Q3 2026 earnings — date not yet formally confirmed (alt. estimate Nov 4).
Structural	Global uranium supply deficit from AI-driven electrification demand + a decade of underinvestment.

BULL CASE

✓ Purest upstream fuel-cycle exposure of the three; FY2025 net income +243% YoY shows real operating leverage emerging.

✓ Westinghouse stake is a legitimate, not-fully-priced-in embedded-value catalyst.

BEAR CASE / RISKS

⚠️ Highest volatility of the three and a demanding valuation (147x+ TTM P/E) with little room for further misses.

⚠️ Two of the last four quarters were significant EPS misses — a recurring pattern, not a one-off; next earnings date unconfirmed.

Sources: MarketChameleon, StockAnalysis.com, MarketBeat, Yahoo Finance — retrieved 2026-10-01.

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CONSOLIDATED CATALYST CALENDAR — Oct 2026 through Q1 2027

Date	Ticker	Event	Type	Confirmed?
2026-10-01	**ALL**	Report-date valuation snapshot; all fundamentals in this workbook as-of this date.	*Reference*	**N/A**
2026-10-30	**CCJ**	Q3 2026 earnings (estimated — not formally confirmed; alt. tracker estimate Nov 4).	*Earnings*	**Estimated**
2026-11-06	**VST**	Q3 2026 earnings. First full quarter with potential Cogentrix commentary.	*Earnings*	**Confirmed**
2026-11-09	**CEG**	Q3 2026 earnings. Consensus EPS $3.65–$3.67.	*Earnings*	**Confirmed**
2026 H2 (TBD)	**VST**	Cogentrix gas portfolio (5,500 MW) acquisition targeted to close.	*M&A / Closing*	**Pending**
2026 (ongoing)	**CEG**	Crane Clean Energy Center restart progress (FERC/NRC approvals granted).	*Regulatory*	**In progress**
2026 (ongoing)	**CEG**	Calpine integration; Brazos Valley + 4.4GW PJM divestiture to LS Power.	*M&A / Regulatory*	**In progress**
2026 (ongoing)	**CCJ**	Westinghouse IPO speculation — potential embedded-value unlock.	*Corporate action*	**Speculative**
2026 (ongoing)	**ALL**	PJM/ERCOT affordability intervention risk — VST most exposed, CEG moderate, CCJ least.	*Regulatory risk*	**Monitoring**
2026 (ongoing)	**CCJ**	Global uranium structural deficit narrative vs. supply growth.	*Macro / Structural*	**Ongoing**
2027 Q1 (est.)	**ALL**	Q4 2026 earnings across all three — first full read on FY2026 guidance delivery and FY2027 initial guides.	*Earnings*	**Estimated**

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PORTFOLIO RECOMMENDATION — Optimizer & Simulation-Informed

RISK / REWARD SNAPSHOT (live-linked to Returns & Risk)			
Metric	**CEG**	**VST**	**CCJ**
Annualized return	-9.0%	-10.6%	40.2%
Annualized volatility	51.0%	38.7%	53.5%
Sharpe ratio	-0.26	-0.38	0.67
Sortino ratio	-0.41	-0.59	1.49
Max drawdown (real)	-33.9%	-33.8%	-30.0%
Historical VaR 95% (monthly)	-19.43%	-13.79%	-13.49%
Historical CVaR 95% (monthly)	-20.55%	-20.45%	-15.20%

OPTIMIZER OUTPUT (live-linked to Portfolio Optimization)			
Portfolio	**w(CEG)**	**w(VST)**	**w(CCJ)**
Max-Sharpe (optimizer)	0%	0%	100%
Min-Volatility (optimizer)	0%	80%	20%
Research Recommendation	40%	35%	25%

Why we don't just take the optimizer's weights: mean-variance optimization on a 20-observation sample is notoriously unstable — it tends to pile weight into whichever asset happened to have the best trailing Sharpe, which is as much noise as signal at this sample size. The Research Recommendation (40/35/25) deliberately keeps CCJ's weight below both the optimizer's and an equal-weight split, specifically because its earnings unpredictability and valuation are a real, separate risk the optimizer's covariance matrix does not capture. Compare all three rows above before sizing a real position.

12-MONTH SCENARIO FRAMING (from Monte Carlo simulation, Research Reco weights)	
Scenario	**Simulated 12-mo return**
Bear (5th percentile)	**-46.1%**
Below-average (25th pct.)	**-20.1%**
Base / median (50th pct.)	**+5.9%**

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Above-average (75th pct.)	**+38.8%**
Bull (95th percentile)	**+103.3%**

Distribution-level risk: P(loss) = 44.3% over 12 months; P(loss > 10%) = 34.8%; simulated 95% VaR = -46.1%, 95% CVaR = -54.3% — i.e., in the worst 5% of simulated outcomes, the average loss is roughly -54%. This is the honest tail-risk picture that a single 'bear case' price target does not convey.

BOTTOM LINE

All three names screen Buy-rated with meaningful upside to analyst targets after a broad 2025/2026 sector de-rating. Benchmarked against XLE, CEG and VST's betas and information ratios (Benchmark & Factors tab) show how much of their recent weakness is sector-wide vs. idiosyncratic; the factor table shows real, if imprecise, sensitivity to oil, gas, uranium and rates. The optimizer confirms a barbell shape is reasonable but, on this short sample, would overweight whichever name had the best trailing luck — a classic mean-variance pitfall — which is why the Research Recommendation deliberately underweights the

DISCLAIMER: All scenario ranges, optimizer output, and Monte Carlo statistics are illustrative, author-generated analysis for portfolio-construction discussion only — not price targets, recommendations, or investment advice. This is not a research report of a registered broker-dealer or investment adviser. See the Cover tab for full methodology and limitations. Consult a licensed financial advisor and verify all data independently before making any investment decision.

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